

MICHAEL MILOWSKI

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EDUCATION

University of Illinois Urbana-Champaign
Bachelor of Science in Aerospace Engineering

August 2024 – May 2028
GPA 3.76/4.00

EXPERIENCES

NASA Langley Research Center – Space Mission Analysis Branch, Pathways

August 2026 – Present

- Supporting engineering activities for Space Mission Cost Team, contributing to development and analysis of Space Missions to inform leadership's decisions.
- Analyzed past and future space missions' technical details to generate cost analyses and assess feasibility.
- Completed a NASA systems engineering course delivered through Caltech, covering requirements development, technical reviews, risk management, verification, and validation.

Woodward Inc. – Aerospace Electronics Systems NPI engineering Intern

May 2026 – Present

- Worked with a small team to troubleshoot and integrate an electro-mechanical thrust reverser actuation system, doing PCB rework and repair, building and reworking cables and performing Acceptance Test Procedure.
- Reviewed and corrected system requirements for a Military actuation system to ensure alignment with customer specifications.
- Analyzed components and assembly weight for complex electronics housing to improve weight estimation.
- Designed and 3D printed tooling to greatly improve efficiency and accuracy of resolver testing prior to installation.

PROJECTS AND LEADERSHIP

Illinois Space Society – NASA Human Lander Challenge (HuLC)

August 2024 – May 2026

- Analyzed cryogenic propellant transfer methods to improve pipe chilldown efficiency by up to 176% and reduce propellant loss by 65%.
- Conducted literature reviews of 30+ research papers and applied findings to Artemis propellant transfer conditions.
- Presented concept to a NASA Glenn cryogenics engineer and was selected as one of 12 HuLC competition finalists.

NASA L'SPACE – Deputy Project Manager of Resources

January 2025 – May 2025

- Led weekly meetings to coordinate goals, communicate progress, and plan key programmatic elements of the mission, while fostering open communication and problem-solving throughout a diverse multidisciplinary team.
- Managed mission cost, schedule, and risk while using NASA mission-planning tools and life-cycle processes to develop a space mission concept through Preliminary Design Review.

Kappa Theta Tau – Professional Society for Distinguished Engineering Students

September 2024 – Present

- Contributed to the 2025 University of Illinois Engineering Open House in designing, prototyping and completing a project, strengthening my knowledge of the engineering process through collaborating with upperclassmen
- Collaborated with a team of peers to plan, organize, and execute three successful fundraising events raising \$1000+ for charity while developing effective communication and project management skills
- Orchestrated a student STEM program at Champaign Public Library to promote engineering in a fun and informative way by developing and leading a lesson and activity for children in afterschool care

AWARDS

Best Technical Presentation – NASA HuLC Team

Dean's List - University of Illinois Urbana-Champaign

Magna Cum Laude Graduate, GPA: 4.5/4.0 – Lake Zurich High School

Illinois State Scholar

AP Scholar with Distinction – Highest AP award for excellence across 8 exams

SKILLS

CAD & Design Software: AutoDesk Inventor, AutoCAD, Siemens NX, Fusion 360, Creo

Programming Languages: C++, Python, Java

Workspace Tools: Excel, IBM Doors, Jira, Windchill, Microsoft 365